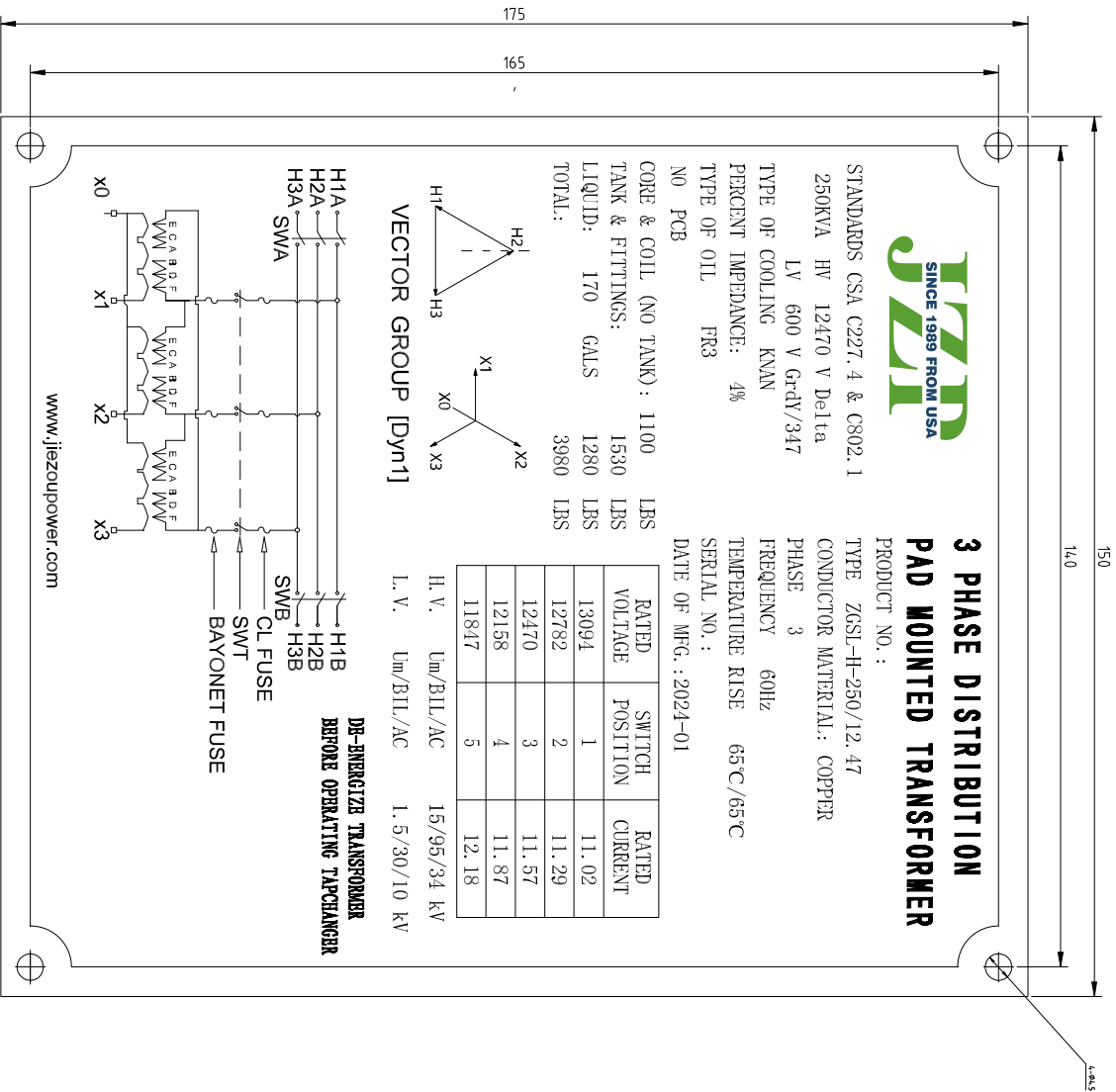


|                              |         |         |  |          |                                        |       |   |  |                    |
|------------------------------|---------|---------|--|----------|----------------------------------------|-------|---|--|--------------------|
| TYPE<br>ZGSL-H-250/12.47/0.6 |         |         |  |          | DWG. NAME<br>General Assembly Drawing  |       |   |  |                    |
| DRAWN                        | CHECKED | CHECKED |  | APPROVED | FIRST ANGLE<br>PROJECTION              | SCALE | 1 |  |                    |
| Mike                         | Jack    | Iris    |  | Sam      | <div>JZP<br/>SINCE 1989 FROM USA</div> |       |   |  | REV.<br>NO.        |
| ZGS250-2401121-01            |         |         |  |          |                                        |       |   |  | <div>△<br/>0</div> |

APPROUVÉ

Signature unitaire de Joel Girardier  
DN : cn=Joel Girardier, o=SGE,  
ou=Estimation & Achats,  
email=Jgirardier@groupesgca  
c=CA  
Date : 2024.02.09 11:03:4 -05'00'

NOTE:  
1.ALL DIMENSIONS ARE IN MILLIMETERS  
2.MATERIAL:STAINLESS STEEL  
3.THICKNESS :1mm  
4.LETTER COLOR: BLACK



|                        |  |               |     |            |                   |         |          |
|------------------------|--|---------------|-----|------------|-------------------|---------|----------|
| TYPE                   |  |               |     | DWG. NAME  |                   |         |          |
| ZGSL-H-250/12.47/0.6   |  |               |     | Name Plate |                   |         |          |
| FIRST ANGLE PROJECTION |  | SCALE         | 1/2 | DRAWN      | CHECKED           | CHECKED | APPROVED |
| 2024-01-21             |  | MikeJack Iris |     |            |                   |         | Sam      |
| CHANGE                 |  | REF. DWG. NO. |     | DWG. NO.   | ZGS250-2401121-01 |         |          |
|                        |  | REV. NO.      |     |            |                   |         |          |

## **Chardon 15kV and 25kV Elbow Arresters**



APPLICATION

The Chardon Elbow Arrester combines metal oxide varistor module in a rubber elbow to provide overvoltage system protection. The Chardon Elbow Arrester housing design meets IEEE 386 standard. It is molded with high quality peroxide-cured insulating and semi-conducting rubber and is fully interchangeable with competitors' products that also meet IEEE standard 386.

Chardon Elbow Arresters are used on underground systems in pad-mounted transformer and entry cabinets, vaults switching devices and other insulations to provide shielded deadfront arrester protection. Installing Chardon Elbow Arrester at both ends of an open point on a loop system and the end of a radial system protect equipment and extend cable life.

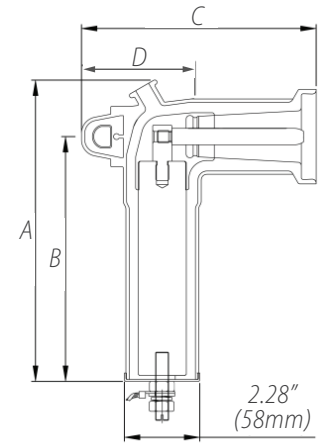
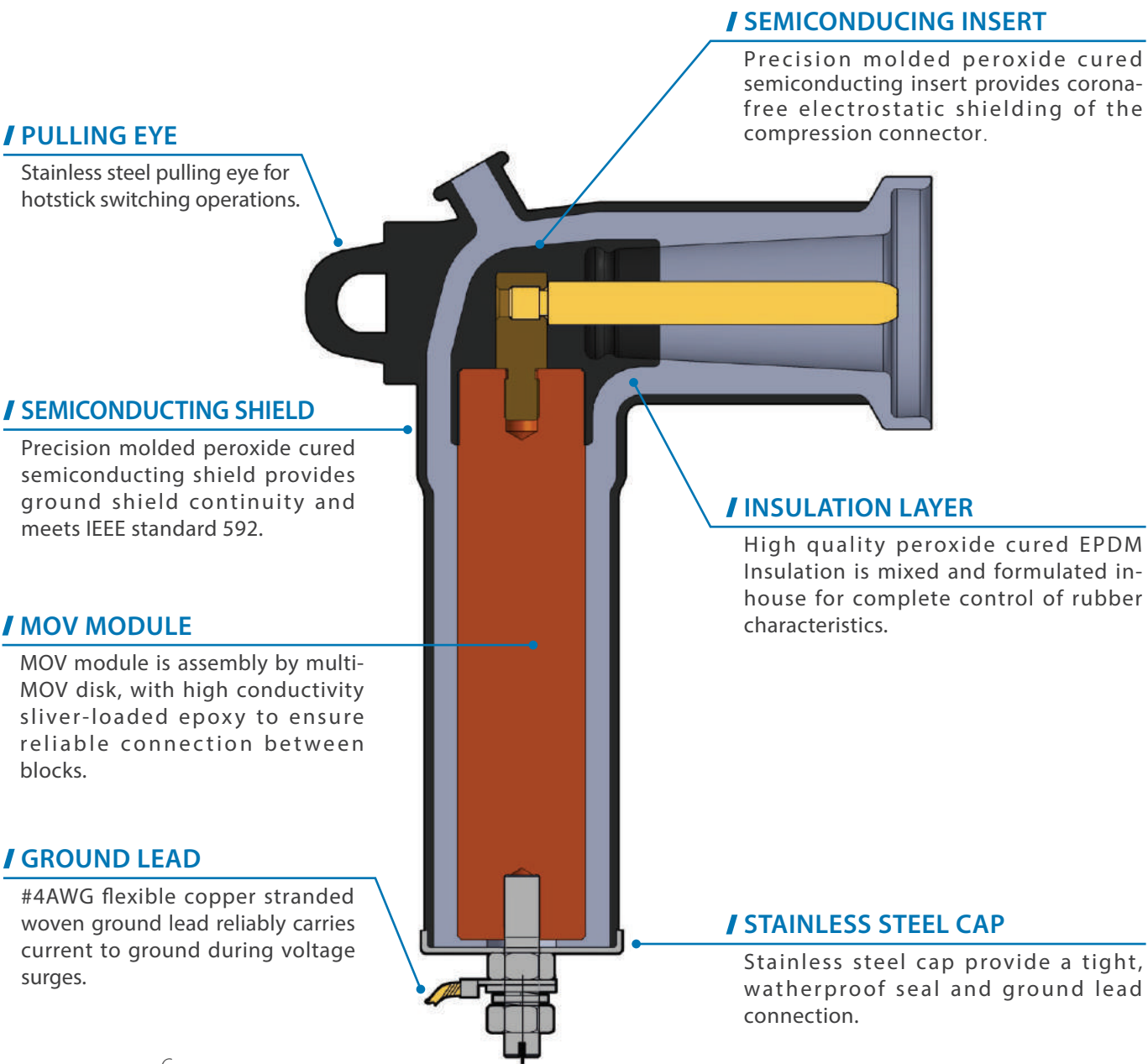
PRODUCTION TEST OF ELBOW ARRESTER

- / Partial Discharge Test
- / Voltage at 1mA
- / Leakage Current
- / Physical Inspection
- / Periodic Dissection
- / Periodic X-ray Analysis

PRODUCTION TEST OF MOV MODULE

- / Partial Discharge Test
- / Voltage at 1mA
- / Leakage Current
- / Batch High-current, Short-duration test
- / Batch Thermal Stability test
- / Batch Aging test
- / Physical Inspection

DETAILED COMPOSITION OF THE CHARDON ELBOW ARRESTERS



| Voltage Class | 15kV  | 25kV  |
|---------------|-------|-------|
| A             | 232mm | 232mm |
| B             | 189mm | 189mm |
| C             | 181mm | 195mm |
| D             | 88mm  | 100mm |

APPROUVÉ

Signature numérique de Joël Grondines  
DN : cn=Joël Grondines,  
o=SGE, ou=Estimation &  
Achat,  
email=jgrondines@groupepsge.  
ca, c=CA  
Date : 2024.02.09 11:09:53  
+05'00'

DETAILED COMPOSITION OF THE  
CHARDON ELBOW ARRESTERS

DETAILED COMPOSITION OF THE  
CHARDON ELBOW ARRESTERS

Table 1

| Electrical Characteristics     |           |                                     |                                                              |      |      |      |      |
|--------------------------------|-----------|-------------------------------------|--------------------------------------------------------------|------|------|------|------|
| Duty Cycle Voltage Rating (kV) | MCOV (kV) | Equivalent Front-of-wave (kV crest) | Maximum Discharge Voltage (kV crest)<br>8/20 µs Current Wave |      |      |      |      |
|                                |           |                                     | 1.5kA                                                        | 3kA  | 5kA  | 10kA | 20kA |
| 3                              | 2.55      | 9.8                                 | 7.7                                                          | 8.0  | 8.5  | 9.5  | 10.6 |
| 6                              | 5.1       | 19.6                                | 15.4                                                         | 16.0 | 17.0 | 18.9 | 21.3 |
| 9                              | 7.65      | 29.3                                | 22.5                                                         | 23.5 | 25.5 | 28.4 | 31.9 |
| 10                             | 8.4       | 32.7                                | 24.9                                                         | 26.0 | 28.0 | 31.1 | 35.0 |
| 12                             | 10.2      | 39.1                                | 30.2                                                         | 31.5 | 34.0 | 37.8 | 42.5 |
| 15                             | 12.7      | 48.9                                | 38.4                                                         | 40.0 | 42.5 | 47.3 | 53.1 |
| 18                             | 15.3      | 58.7                                | 46.0                                                         | 48.0 | 51.0 | 56.7 | 63.8 |
| 21                             | 17.0      | 68.4                                | 53.6                                                         | 56.0 | 59.5 | 66.0 | 74.0 |

Table 2

| System Voltage (kV rms) |                         | Commonly Applied Arrester Duty-cycle(MCOV)<br>Voltage Rating (kV rms) on Distribution System |                               |                                          |
|-------------------------|-------------------------|----------------------------------------------------------------------------------------------|-------------------------------|------------------------------------------|
| Nominal Voltage         | Maximum Voltage Range B | 4-Wire Multigrounded Neutral Wye                                                             | 3-Wire Low Impedance Grounded | Delta and 3-wire High Impedance Grounded |
| 2.4                     | 2.54                    | -                                                                                            | -                             | 3(2.55)                                  |
| 4.16 Y/2.40             | 4.40 Y/2.54             | 3(2.55)                                                                                      | 6(5.1)                        | 6(5.1)                                   |
| 4.26                    | 4.40                    | -                                                                                            | -                             | 6(5.1)                                   |
| 4.80                    | 5.08                    | -                                                                                            | -                             | 6(5.1)                                   |
| 6.90                    | 7.26                    | -                                                                                            | -                             | 9(7.65)                                  |
| 8.32 Y/4.80             | 8.80 Y/5.08             | 6(5.1)                                                                                       | 9(7.65)                       | -                                        |
| 12.00 Y/6.93            | 12.7 Y/7.33             | 9(7.65)                                                                                      | 12(10.2)                      | -                                        |
| 12.47 Y/7.20            | 13.20 Y/7.62            | 9(7.65) or 10(8.4)                                                                           | 15 (12.7)                     | -                                        |
| 13.20 Y/7.62            | 13.97 Y/8.07            | 10(8.4)                                                                                      | 15 (12.7)                     | -                                        |
| 13.80 Y/7.97            | 14.52 Y/8.388           | 10(8.4) and 12(10.2)                                                                         | 15 (12.7)                     | -                                        |
| 13.80                   | 14.52                   | -                                                                                            | -                             | 18(15.3)                                 |
| 20.76 Y/12.00           | 22.00 Y/12.70           | 15 (12.7)                                                                                    | -                             | --                                       |
| 22.86 Y/12.00           | 22.00 Y/12.70           | 15 (12.7)                                                                                    | -                             |                                          |
| 24.94 Y/14.40           | 26.40 Y/15.24           | 18 (15.3)                                                                                    | -                             | -                                        |
| 27.60 Y/15.935          | 29.255 Y/16.89          | 21 (17.0)                                                                                    | -                             | -                                        |

PROTECTIVE CHARACTERISTICS

The protective characteristics of the elbow arrester are shown in Table 1.

GENERAL APPLICATION RECOMMENDATION

The rating of an arrester is the maximum power frequency line-to-ground voltage at which the arrester is designed to pass an operating duty-cycle test. Table 2 provides a general application guide for the selection of the proper arrester rating for a given system voltage and system grounding configuration as outlined in the IEEE Std C62.22 standard application guide.

To ensure that the arrester ratings will not be exceeded, CHARDON GROUP application engineers are available to make recommendations.

PERFORMANCE TEST CHARACTERISTICS

The Elbow Arrester consistently withstands the following design test:

- / Duty Cycle Test
- / High-current, Short-duration Discharge test
- / Low-Current, Long-Duration Discharge test

ORDERING INFORMATION

| IEEE Std 386 Standard Interface | Duty Cycle (kV) | MCOV Rating (kV) | Catalog Number |
|---------------------------------|-----------------|------------------|----------------|
| 15kV                            | 3 kV            | 2.55 kV          | 15-LEA3        |
|                                 | 10 kV           | 8.4 kV           | 15-LEA10       |
|                                 | 12 kV           | 10.2 kV          | 15-LEA12       |
|                                 | 15 kV           | 12.7 kV          | 15-LEA15       |
|                                 | 15 kV           | 12.7 kV          | 25-LEA15       |
|                                 | 18 kV           | 15.3 kV          | 15-LEA18       |
|                                 | 21 kV           | 17.0 kV          | 15-LEA21       |
| 25kV                            | 10 kV           | 8.4 kV           | 25-LEA10       |
|                                 | 12 kV           | 10.2 kV          | 25-LEA12       |
|                                 | 18 kV           | 15.3 kV          | 25-LEA18       |
|                                 | 21 kV           | 17.0 kV          | 25-LEA21       |



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